Cyclistic Bike Share Data Analysis – 2020 Case Study

Google Data Analytics Portfolio Project

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Tools Used: SQL (Google BigQuery), Excel, Tableau, R Markdown

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Executive Summary

This analysis examines Cyclistic's 2020 bike-share data to identify behavioral differences between casual riders and annual members and evaluate opportunities to increase membership conversions.

The dataset contains **3.47 million rides** across twelve months. After cleaning and integrating the data using SQL in Google BigQuery, exploratory analysis was conducted to evaluate ride frequency, ride duration, temporal patterns, and station-level usage.

The analysis reveals clear behavioral differences between the two rider groups. Annual members demonstrate consistent weekday commuting patterns with shorter ride durations, while casual riders primarily use the service for longer recreational trips, particularly during weekends and summer months.

Geographic analysis further shows that casual riders cluster around tourist and recreational areas, while members concentrate near business districts and transit corridors.

Based on these findings, targeted marketing campaigns focused on high-traffic recreational stations and peak seasonal demand may help convert casual riders into annual members.

Introduction

Cyclistic is a Chicago-based bike-share company that offers single-ride passes, day passes, and annual memberships. The company's marketing team aims to increase annual memberships by converting casual riders into long-term subscribers.

This case study analyzes Cyclistic's 2020 ride data to identify behavioral differences between casual riders and annual members and provide data-driven recommendations to support membership conversion strategies.

Analysis Workflow

The analysis followed a structured workflow including data cleaning, validation, feature engineering, exploratory analysis, and business insight generation. SQL was used for data transformation and aggregation in Google BigQuery, while Tableau was used to visualize behavioral patterns between casual riders and annual members.

1. Data acquisition and review of the Cyclistic 2020 trip dataset
2. Data cleaning and validation
3. Feature engineering of ride duration and time-based fields
4. Integration of monthly datasets in Google BigQuery
5. Exploratory data analysis of rider behavior patterns
6. Visualization of insights using Tableau

7. Development of business insights and recommendations

Data Sources

The dataset used in this analysis consists of Cyclistic's publicly available 2020 bike-share trip data. The dataset includes ride-level information such as:

• Ride start and end timestamps
• Start and end station names
• Rider type (member or casual)
• Ride duration

The dataset contained over **3.47 million ride records** across all twelve months of 2020.

Data Cleaning and Preparation

Before analysis, each monthly dataset was reviewed to ensure structural consistency and realistic ride behavior.

-Ride Duration Validation

To remove invalid or unrealistic ride records, the following filters were applied using Excel:

• Rides with duration ≤ 0 minutes were removed (system errors or corrupted entries)
• Rides exceeding 1440 minutes (24 hours) were removed as likely logging errors
• Rides lasting at least 1 minute were retained as valid user activity

These steps removed approximately 0.8–1.2% of records per month, representing expected real-world data noise.

-Data Standardization

To ensure consistency across all datasets; using Excel:

• Leading and trailing whitespace was trimmed from station names
• Blank station names were replaced with "Unknown" to preserve valid rides
• Rider type values were standardized using LOWER(TRIM())
• Ride length calculations were converted to static values after validation

Feature Engineering

Additional fields were created to support behavioral analysis; using Excel:

• `ride_length_minutes` – ride duration in minutes
• `day_of_week` – weekday extracted from ride timestamp
• `start_ride_hour` – ride start hour
• `time_of_day` – categorized as Morning, Afternoon, or Evening

These derived features enabled deeper analysis of riding patterns across user types.

Data Integration and Validation

All twelve cleaned monthly datasets were uploaded to **Google BigQuery** and combined into a master table using `UNION ALL`.

The final dataset contained **3,476,988 rides**, matching the sum of all individual monthly datasets and confirming that no data loss occurred during integration.

Additional validation steps included:

- Verifying column data types across all tables
- Checking timestamp consistency
- Removing records with missing key fields (0.1% of total records)

A clean analysis-ready dataset was then created to support downstream queries and visualization.

SQL Code used:

```
CREATE OR REPLACE TABLE `cyclistic-2020-488102.cyclistic.2020_all_rides` AS
SELECT * FROM `cyclistic-2020-488102.cyclistic.q1_2020_jan_clean`
UNION ALL
SELECT * FROM `cyclistic-2020-488102.cyclistic.q1_2020_feb_clean`
UNION ALL
SELECT * FROM `cyclistic-2020-488102.cyclistic.q1_2020_mar_clean`  --
additional months omitted for brevity
```

Exploratory Data Analysis

Ride Volume by User Type

Members accounted for approximately 61% of total rides (2.13 million)**,** while casual riders represented 39% (1.34 million)**.**

Although members generate the majority of usage, the substantial number of casual riders indicates a strong opportunity for membership conversion.

SQL Used:

```
SELECT   member_casual,

  COUNT(*) AS total_rides

FROM `cyclistic-2020-488102.cyclistic.2020_all_rides_analysis`

GROUP BY member_casual

ORDER BY total_rides DESC;
```

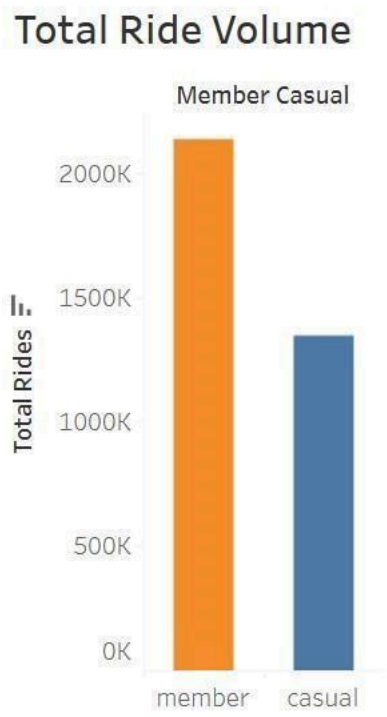

Figure 1 – Ride Volume by User Type
Members account for the majority of total rides, indicating consistent reliance on Cyclistic bikes for routine transportation.

Ride Duration by User Type

Casual riders averaged 39.26 minutes per ride, compared to 15.83 minutes for members**.**

Median ride length further reinforces this pattern:

• Casual riders: 22 minutes
• Members: 11 minutes

These results suggest that casual riders primarily use bikes for longer recreational trips, while members rely on bikes for shorter commuting trips.

SQL Used:

```
SELECT

member_casual,

  ROUND(AVG(ride_length_minutes), 2) AS avg_ride_length_minutes

FROM `cyclistic-2020-488102.cyclistic.2020_all_rides_analysis`
GROUP BY member_casual;
```

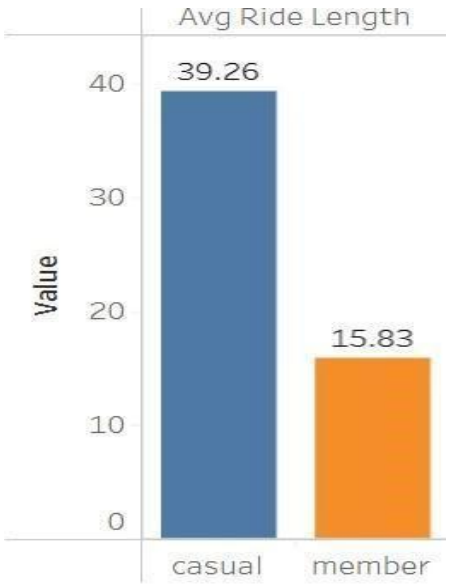

Figure 2 – Ride Duration by User Type

Casual riders exhibit significantly longer average ride durations than members, suggesting recreational rather than commuting usage patterns.

Usage by Day of Week

Members demonstrate significantly higher ride volume during weekdays, indicating commuter-driven usage.

For example:

- Tuesday: 292,501 member rides vs 130,871 casual rides • Thursday: 306,110 member rides vs 155,003 casual rides On weekends, the gap narrows substantially:

- Saturday: 293,771 member rides vs 303,996 casual rides
- Sunday: 253,372 member rides vs 246,976 casual rides

This suggests that casual riders are more likely using Cyclistic bikes for leisure activities**.**

SQL Used:

```
SELECT
 CASE
  WHEN day_of_week IN ('1','2','3','4','5','6','7') THEN
   CASE day_of_week
    WHEN '1' THEN 'Sunday'
    WHEN '2' THEN 'Monday'
    WHEN '3' THEN 'Tuesday'
```

```
      WHEN '4' THEN 'Wednesday'
      WHEN '5' THEN 'Thursday'
      WHEN '6' THEN 'Friday'
      WHEN '7' THEN 'Saturday'
    END
  ELSE day_of_week   END
AS day_name,
member_casual,
 COUNT(*) AS total_rides
FROM `cyclistic-2020-488102.cyclistic.2020_all_rides`
WHERE day_of_week IS NOT NULL
GROUP BY day_name, member_casual
ORDER BY day_name;
```

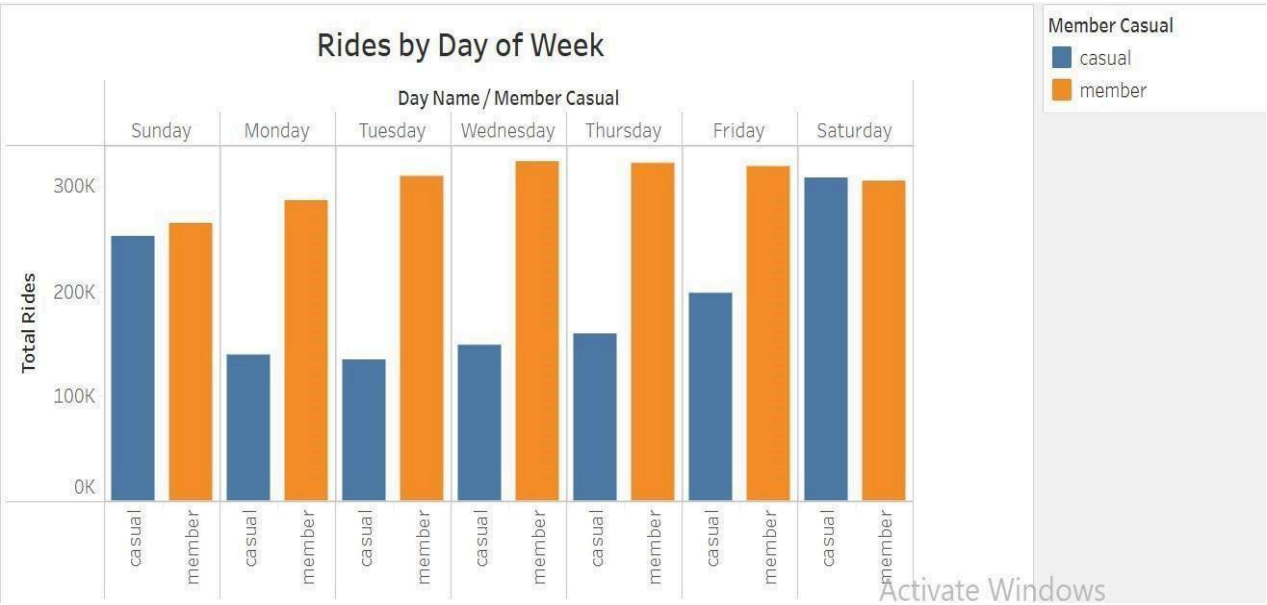

Figure 3 – Ride Volume by Day of Week

Members demonstrate higher weekday ride volume, while casual riders show increased activity on weekends.

Time-of-Day Patterns

Members exhibit clear commuting patterns, with ride peaks during:

• Morning commute hours (7–9 AM)
• Evening commute hours (4–6 PM)

Casual riders demonstrate stronger afternoon usage, indicating recreational or flexible ride behavior.

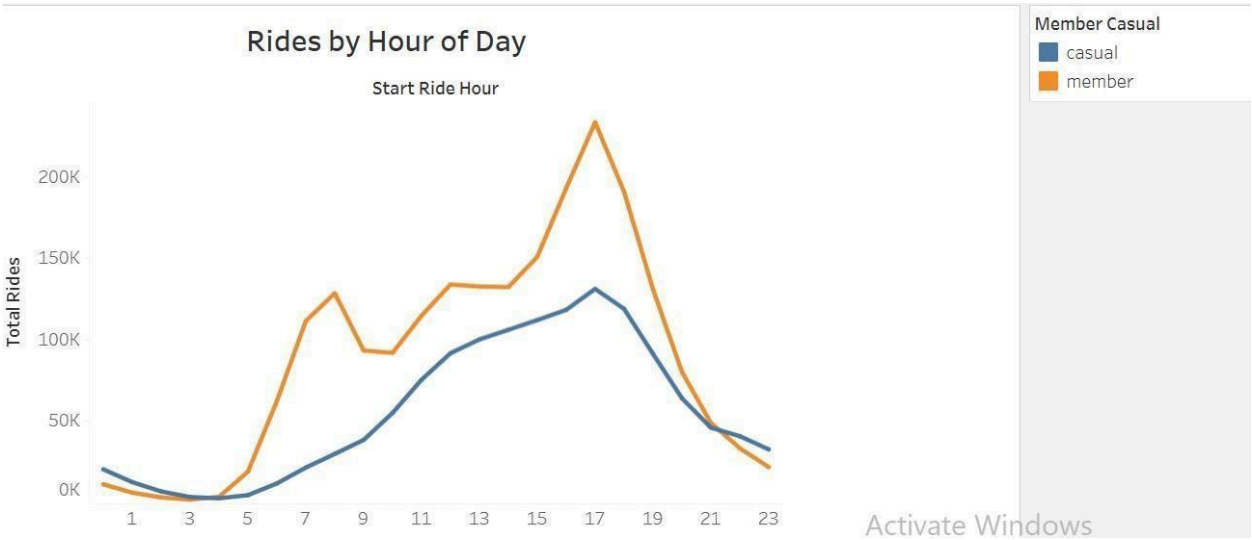

Figure 4 – Time-of-Day Ride Patterns

Member rides peak during morning and evening commute hours, while casual riders show stronger afternoon riding activity.

Monthly & Seasonal Usage Trends

Casual ridership shows strong seasonal variation.

Winter months show minimal casual usage:

• January: 7,785 casual rides
• February: 12,869 casual rides

Casual ridership increases significantly during warmer months:

• July: 265,411 casual rides
• August: 283,034 casual rides

Member usage remains relatively stable throughout the year, reinforcing the interpretation that members rely on Cyclistic bikes for routine transportation.

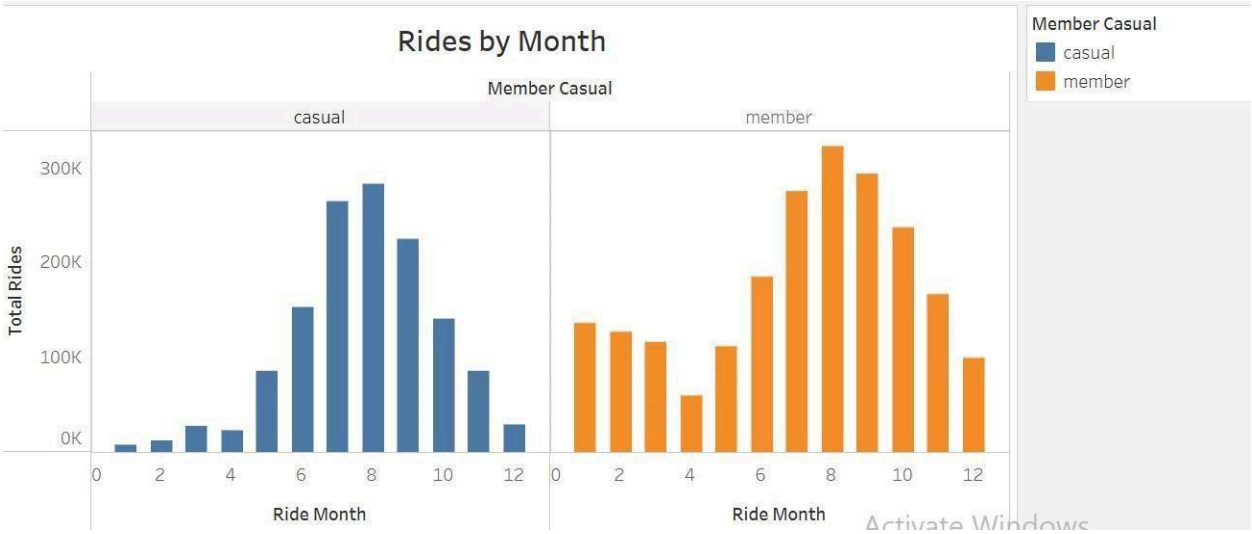

Figure 5 – Monthly Ride Trends
Casual ridership increases significantly during summer months, while member usage remains relatively stable throughout the year.

Top Start Stations

Casual riders are concentrated around recreational and tourist locations such as:

- Streeter Dr & Grand Ave
- Lake Shore Dr & Monroe St
- Millennium Park
- Theater on the Lake

Member riders are concentrated near business and commuter areas, including:

- Clark St & Elm St
- Kingsbury St & Kinzie St
- St. Clair St & Erie St
- Clinton St & Madison St

This geographic segmentation further supports the difference between recreational and commuter riding patterns.

Ride Length Distribution & Median Analysis

Ride-length distribution reveals clear behavioral differences between rider types beyond simple averages.

Approximately 76% of member rides fall under 20 minutes, confirming consistent short-duration commuting patterns. In contrast, casual riders exhibit a significantly heavier long-duration distribution, with over 15% of rides exceeding 60 minutes**.**

This distributional difference reinforces the interpretation that members primarily use Cyclistic bikes for routine transportation, while casual riders are more likely engaging in extended recreational trips**.**

SQL Used:

```
-- Ride length distribution by user type (count + percent)

WITH binned AS (

SELECT

member_casual,

  CASE

    WHEN ride_length_minutes BETWEEN 1 AND 10 THEN '01–10 min'

    WHEN ride_length_minutes BETWEEN 11 AND 20 THEN '11–20 min'

    WHEN ride_length_minutes BETWEEN 21 AND 30 THEN '21–30 min'

    WHEN ride_length_minutes BETWEEN 31 AND 45 THEN '31–45 min'

WHEN ride_length_minutes BETWEEN 46 AND 60 THEN '46–60 min'

    WHEN ride_length_minutes > 60 THEN '60+ min'

    ELSE 'Other/Invalid'

  END AS ride_length_bin

 FROM `cyclistic-2020-488102.cyclistic.2020_all_rides_analysis`

)

SELECT   member_casual,

ride_length_bin,

 COUNT(*) AS rides,

 ROUND(

  100 * COUNT(*) / SUM(COUNT(*)) OVER (PARTITION BY member_casual),

  2

 ) AS pct_within_user_type
FROM binned
WHERE ride_length_bin != 'Other/Invalid'
```

```
GROUP BY member_casual, ride_length_bin
ORDER BY   member_casual,
 CASE ride_length_bin
   WHEN '01–10 min' THEN 1
   WHEN '11–20 min' THEN 2
   WHEN '21–30 min' THEN 3
   WHEN '31–45 min' THEN 4
   WHEN '46–60 min' THEN 5
   WHEN '60+ min' THEN 6
   ELSE 7
 END;
```

Median ride length further reinforces this behavioral difference. The typical casual ride lasts approximately 22 minutes, compared to about 11 minutes for annual members. Because median values are less sensitive to extreme outliers than averages, this confirms that casual riders consistently take longer trips even when controlling for unusually long rides.

SQL Used:

```
SELECT
member_casual,
 APPROX_QUANTILES(ride_length_minutes, 100)[OFFSET(50)] AS median_ride_length_minutes
FROM `cyclistic-2020-488102.cyclistic.2020_all_rides_analysis`
GROUP BY member_casual;
```

Data Limitations

This analysis is based solely on trip-level ride data and does not include demographic or behavioral information about individual riders. As a result, it is not possible to determine whether casual riders are local residents, tourists, or infrequent users, and rider motivations must be inferred from observed usage patterns.

Additionally, the dataset reflects travel behavior during 2020, a year significantly affected by the COVID-19 pandemic, which may have influenced commuting patterns, seasonal usage, and overall ridership levels.

Finally, station-level analysis assumes that start locations reflect rider intent; however, factors such as bike availability, station proximity, or temporary operational changes may also influence station selection.

Despite these limitations, the dataset provides strong behavioral signals that highlight meaningful differences between casual riders and annual members.

Key Findings

1. Members account for 61% of total rides and exhibit consistent year-round usage.
2. Casual riders take significantly longer rides than members.
3. Members dominate weekday commuting hours, while casual riders peak on weekends.
4. Casual ridership increases dramatically during summer months.
5. Geographic patterns show casual riders concentrated near tourist and recreational areas, while members cluster around business districts and transit corridors.

Business Recommendations

1. Launch targeted seasonal marketing campaigns during summer months when casual ridership peaks.
2. Deploy QR-code membership promotions at high-traffic recreational stations to capture impulse interest from leisure riders.
3. Introduce limited-time “Weekend-to-Membership” upgrade incentives to encourage frequent casual riders to transition to annual memberships.
4. Promote commuter benefits near business districts and transit hubs, emphasizing cost savings and convenience.
5. Focus conversion campaigns during peak months when casual and member ride volumes converge.

Conclusion

This analysis identified clear behavioral differences between casual riders and annual members. Members primarily use Cyclistic bikes for routine transportation and commuting, resulting in shorter rides and higher weekday activity. Casual riders demonstrate longer ride durations, stronger weekend usage, and significant seasonal variation.

These insights suggest that casual riders represent a large pool of potential membership conversions, particularly during summer months and at recreational station locations. Targeted marketing strategies that focus on these behavioral patterns may help Cyclistic expand its annual membership base and improve long-term rider retention.

Additional SQL Queries

-Monthly / Seasonal Trend Query

SQL Used:

```
SELECT
  EXTRACT(MONTH FROM started_at) AS month_num,
FORMAT_TIMESTAMP('%B', started_at) AS month,   member_casual,
  COUNT(*) AS total_rides
FROM `cyclistic-2020-488102.cyclistic.2020_all_rides_analysis`
GROUP BY month_num, month, member_casual
ORDER BY month_num, member_casual;
```

-Time-of-Day Query

SQL Used:

```
SELECT
start_ride_hour,
member_casual,
  COUNT(*) AS total_rides
FROM `cyclistic-2020-488102.cyclistic.2020_all_rides_analysis`
GROUP BY start_ride_hour, member_casual
ORDER BY start_ride_hour;
```

-Top Start Stations – Casual Riders

SQL Used:

```
SELECT
start_station_name,
  COUNT(*) AS total_rides
FROM `cyclistic-2020-488102.cyclistic.2020_all_rides_analysis`
WHERE member_casual = 'casual'
  AND start_station_name IS NOT NULL
  AND start_station_name != 'Unknown'
GROUP BY start_station_name
```

```
ORDER BY total_rides DESC
LIMIT 10;
```

Top Start Stations – Member Riders

SQL Used:

```
 SELECT   start_station_name,
  COUNT(*) AS total_rides
FROM `cyclistic-2020-488102.cyclistic.2020_all_rides_analysis`
WHERE member_casual = 'member'
  AND start_station_name IS NOT NULL
  AND start_station_name != 'Unknown'
GROUP BY start_station_name
ORDER BY total_rides DESC
LIMIT 10;
```